Note: In this Exam Preparation, we'll cover just the basic skills you need to know in this unit.
For more challenging problems, find out in **Final Exam Preparation** which is already in MCV.

Least Square Problem

1. Given a set of data points $(x, y) \in \{(0,0), (0,1), (1,0)\}$, use the least square method to find the line that best fit such data points in the form of $y = mx + c$.

Eigenvector/Eigenvalue and Eigen decomposition

2. Find all eigenvalues along with their corresponding eigenvectors of the following matrix $\mathbf{A}$. Also, diagonalize the matrix $\mathbf{A}$ using eigen decomposition.

   a. $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$

b. $\mathbf{A} = \begin{bmatrix} 2 & 0 & 0 \\ 3 & -1 & 0 \\ 1 & -2 & 3 \end{bmatrix}$

3. Find an **orthogonal diagonalization** of $\boldsymbol{A} = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix}$

Singular Value Decomposition (SVD)

4. Given matrix $\mathbf{A}$ in each sub-question below, find the following:
   1) SVD of $\mathbf{A}$.
   2) basis of the following spaces: row($\mathbf{A}$), col($\mathbf{A}$), nul($\mathbf{A}$), and nul($\mathbf{A}^{\mathrm{T}}$)
   3) rank($\mathbf{A}$) and nullity($\mathbf{A}$)

   a. $\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ (Actually, this is the same as the example in the video, try find it by yourself.)

b. $\mathbf{A} = \begin{bmatrix} 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ -1 & 1 & -1 & 1 \end{bmatrix}$

Pseudo-Inverse

5. Find $\mathbf{A}^+$ of the given matrix in following question. (Note that it is the same as in question 4, so you can reuse the matrix you have found from the previous question.)

   a. $\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

   b. $\mathbf{A} = \begin{bmatrix} 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ -1 & 1 & -1 & 1 \end{bmatrix}$